

Evaluation of Accuracy and Algorithm Specialization

The algorithms demonstrated varying effectiveness depending on the type of change point. It is worth noting that the maximum value for the metrics is 1.0, and values above 0.7 in streaming (online) analysis tasks without prior training are considered a high indicator (the algorithm makes few errors).

1.1 Results on the “Mean Shift” (Mean) Dataset

In this time series, the mean line of the signal shifts upwards.

Table 1: Results for the dataset: Mean Shift (Mean)

Algorithm	Abrupt Change			Smooth Change		
	AUPRC	F1 (3σ)	Delay	AUPRC	F1 (3σ)	Delay
Tsallis ($q = 2$)	0.943	0.802	8.0	0.851	0.750	40.1
Renyi ($q = 2$)	0.791	0.795	7.6	0.858	0.750	37.5
Shannon	0.705	0.788	4.2	0.817	0.758	34.0
Approximate	0.621	0.592	19.7	0.640	0.571	45.2
Dispersion	0.450	0.773	14.9	0.460	0.765	31.8
Slope	0.071	0.058	26.0	0.041	0.113	42.3
Bubble	0.048	0.095	45.0	0.108	0.198	50.5

1.2 Results on the “Variance Change” (Variance) Dataset

Here, an abrupt and smooth increase in signal noise was simulated.

Table 2: Results for the dataset: Variance Change (Var)

Algorithm	Abrupt Change			Smooth Change		
	AUPRC	F1 (3σ)	Delay	AUPRC	F1 (3σ)	Delay
Approximate	0.938	0.844	25.5	0.936	0.851	43.1
Tsallis ($q = 2$)	0.743	0.830	14.2	0.775	0.780	39.6
Slope	0.660	0.830	44.2	0.640	0.667	69.5
Dispersion	0.620	0.809	41.5	0.345	0.726	64.2
Renyi ($q = 2$)	0.515	0.857	12.5	0.554	0.788	36.8
Shannon	0.506	0.830	8.0	0.530	0.788	31.7

1.3 Results on the “Autoregression Shift” (AR) Dataset

In this series, the signal amplitude remains constant, but the hidden sequence order of points changes.

Table 3: Results for the dataset: Autoregression Shift (AR)

Algorithm	Abrupt Change			Smooth Change		
	AUPRC	F1 (3σ)	Delay	AUPRC	F1 (3σ)	Delay
Slope	0.825	0.742	21.8	0.322	0.450	74.3
Permutation	0.644	0.630	68.3	0.443	0.450	54.8
Bubble	0.477	0.758	30.2	0.836	0.726	53.7
Renyi ($q = 2$)	0.145	0.276	69.5	0.555	0.592	56.3
Shannon	0.131	0.261	66.4	0.504	0.592	57.2
Tsallis ($q = 2$)	0.080	0.113	37.2	0.670	0.658	52.5

1.4 Results on the “Frequency Change” (Quasi-Periodic) Dataset

The series is a sinusoid whose base frequency increases abruptly or smoothly.

Table 4: Results for the dataset: Frequency Change (Quasi)

Algorithm	Abrupt Change			Smooth Change		
	AUPRC	F1 (3σ)	Delay	AUPRC	F1 (3σ)	Delay
Dispersion	0.993	0.974	48.4	0.896	0.830	66.8
Tsallis ($q = 2$)	0.989	0.925	14.4	0.732	0.601	51.2
Renyi ($q = 2$)	0.975	0.925	15.0	0.835	0.684	50.7
Bubble	0.811	0.857	52.2	0.805	0.765	81.4
Shannon	0.353	0.319	18.6	0.016	0.039	51.5

1.5 Results on the “Appearance of Outliers” (Cauchy) Dataset

Table 5: Results for the dataset: Appearance of Outliers (Cauchy)

Algorithm	Abrupt Change			Smooth Change		
	AUPRC	F1 (3σ)	Delay	AUPRC	F1 (3σ)	Delay
Tsallis ($q = 2$)	0.871	0.795	40.5	0.684	0.601	65.0
Renyi ($q = 2$)	0.856	0.809	36.6	0.771	0.675	65.5
Approximate	0.826	0.758	46.0	0.600	0.518	72.8
Shannon	0.805	0.809	29.7	0.816	0.718	60.2
Dispersion	0.164	0.198	53.0	0.097	0.182	45.5

Experiments confirmed the strict specialization of the methods. Parametric entropies (Tsallis, Renyi, Shannon) are effective for amplitude shifts and outliers (AUPRC > 0.85), as they detect changes in the probability density function but do not react to frequency changes. At the same time, generalized methods with the parameter $q = 2$ outperform classic Shannon due to a higher sensitivity to rare deviations, while the KLD metric showed a tendency for false alarms on real-world data. Pattern-based methods (Slope, Dispersion) successfully detect structural and frequency changes (AUPRC > 0.82), as they analyze the sequence order of points but ignore baseline shifts. Similar acting algorithms Permutation and Bubble showed comparable results, but yielded to some algorithms due to a significantly larger detection delay. Complexity algorithms (Approximate, Sample) proved to be the most accurate in detecting variance and noise level changes (AUPRC > 0.93), as they directly assess the drop in series regularity. However, their high computational complexity increases processing time by 6 times compared to parametric counterparts, which limits their application in high-load systems.

1.6 Computational Performance

Table 6 presents the average execution time of each algorithm when analyzing a single time series consisting of 1000 data points.

Table 6: Average processing time of a single time series ($N = 1000$)

Algorithm	Execution Time (ms)
Tsallis ($q = 2$)	10.3
Renyi ($q = 2$)	10.3
Shannon	10.3
Permutation	42.3
Slope	45.5
Dispersion	46.1
Sample	59.5
Approximate	67.3

1.7 Validation on Historical Data

Table 7: Detection results on the UK Coal Mining Disasters dataset

Algorithm	True Point	Detected Point	Delay	Status
Tsallis ($q=2$)	39	40	1	TP (Success)
Renyi ($q=2$)	39	40	1	TP (Success)
Shannon	39	41	2	TP (Success)
Dispersion	39	43	4	TP (Success)
KLD	39	30	–	FP (False Alarm)
Approximate	39	50	–	FN (Too late)
Sample	39	51	–	FN (Too late)
Slope	39	70	–	FN (Too late)
Permutation	39	86	–	FN (Too late)
Bubble	39	Not found	–	FN (Not found)

The algorithms were tested on the real dataset “UK Coal Mining Disasters”. The results completely matched the conclusions drawn from synthetic data (Mean Shift): parametric algorithms Renyi and Tsallis detected the drop in accident rates with a minimal delay of 1 year and zero false alarms. Pattern-based algorithms missed the change point, as the random nature of the accidents did not change. This proves the readiness of the developed library for practical application on real data.